CS-414
AI

Artificial Intelligence

- Incremental learning in Multi-dimensions
- Trained on the data on WWW.
- Will Surpass Humans.
- Biggest Existential Threat

Early Era Drawbacks:-
- Limited Data Storage
- Limited Computational Power

Two Views of AI:
- Agents → Intelligent Robots
- Tools → To benefit humans.

Machines → narrow task, millions of examples
Humans → dense task, very few examples.

PARADIGM:
- Modeling → pick-up useful data.
- Inference → decision making.
- Learning → using data.

ML:-

Data → Model.
Requires leap of faith → Generalization.

- Reflex-based Models → based on how it (low level intelligence) has trained on data.
  Can't change later.
  No back-tracking.
- State-based Models → Games.
- Variables-based
- Logic-based. (high level intelligence)

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Reflex Models

Machine Learning → Supervised (Target Output provided)
→ Unsupervised (" not provided)
→ Reinforcement

SUPERVISED LEARNING:-

- Linear Predictors → Classification Problem (Discrete) {+1, -1}
  → Regression Problem (Continuous) {ℝ}

  Target Value

e.g.

- Spam Classification (Prediction) :-

  Input : x : email txt

  Output : $y \in$ {Spam, Not Spam}

  Objective : Obtain a predictor f.

Classification Problem → Structured Predictions
→ 2 Classes → Binary
→ More Than 2 → Multi-Class
→ Ranking

~~e.g. Ranking~~

Data

e.g. (x, y)

x ↳ Input

y ↳ target output

D_{train} → $\boxed{\text{learner}}$ → $\boxed{f}$ (x → f → y)

$\boxed{\text{Modeling}}$

Inference

Learning

imp and Relevent features from data.

f is based on Feature Extractor.

Depending on Problem, feature's may vary that feature extractor extracts.

$$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0.9 \end{bmatrix}$$

feature vector

Linear Predictor.

5 features $\longrightarrow$ weight vector size (5)

(once against each.

Weight vector-

$$w \in \mathbb{R}^d \text{ if } \phi(x) \in \mathbb{R}^d$$

$\phi(x)$ depends on x

w does not depend on x!

Incase of binary features,

+ive weight and $\phi(x) = 1$

$\Rightarrow$ +ive class.

-ive w and $\phi(x)=1 \Rightarrow$ -ive class.

Score: $w \cdot \phi(x) = \sum_{j=1}^{d} w_j \cdot \phi_j(x)$

f

confidence deta hai ur your value.

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Reflex Models

$x \rightarrow \boxed{f} \rightarrow y$

feature Extractor (ϕ)

⇒ Manual feature Extraction
↳ may extract irrelavant f.
↳ predictor might not work properly.

$x \rightarrow \phi \rightarrow \phi(x) \in \mathbb{R}^d$

real no. — $\mathbb{R}$; no. of features — d

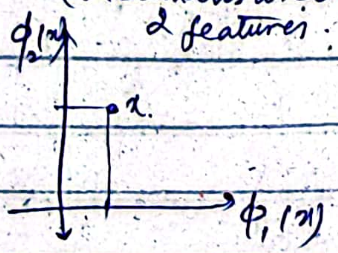
⇒ Most Impo Task
↳ Relavent f extraction.

feature = (f.name, f.value)

feature vector:-

$$\phi(x) = \begin{bmatrix} \phi_1(x) \\ \phi_2(x) \\ \vdots \\ \phi_n(x) \end{bmatrix}$$

if d=2
(2 dimensional)
2 features



Weight Vector:-

$$w = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$$

No. of w = No. of f.
dimensions same too.

↳ does not depend on x.

Score:-

+ive value ⇒ +ive class (+1)
-ive value ⇒ -ive class (-1)

greater the magnitude → greater the confidence that you're right in your prediction.

feature vector → binary 1
weight → +ive
} influence towards +ive Class.

Regression
Classification

Linear Predictor :-

$\phi(x) \in \mathbb{R}^d$, $w \in \mathbb{R}^d$ → Determine f.

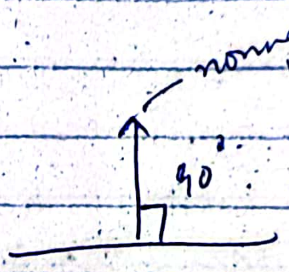
- **Linear/Binary Classifier :-**

$$f_w(x) = \text{sign}(w \cdot \phi(x))$$

$$= \begin{cases} +1 & \text{if } w \cdot \phi(x) > 0 \\ -1 & \text{if } w \cdot \phi(x) < 0 \\ ? & \text{otherwise} \end{cases}$$

indecisive
(can't decide)

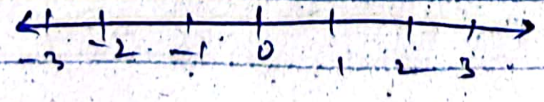
- Geometric Intuition :-



→ f_w is a classifier
defines a hyperplane with normal vector w.

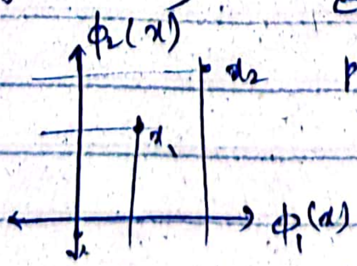
$\phi(x) \in \mathbb{R}$ only 1 feature.

$x_1 = \phi_1(x)$ all values lie on one axis.
$x_2 = \phi_1(x)$



$\phi(x) \in \mathbb{R}^2$ (2 features)

$x_1 = (\phi_1(x), \phi_2(x))$
$x_2 = (\phi_1(x), \phi_2(x))$



Hyperplane

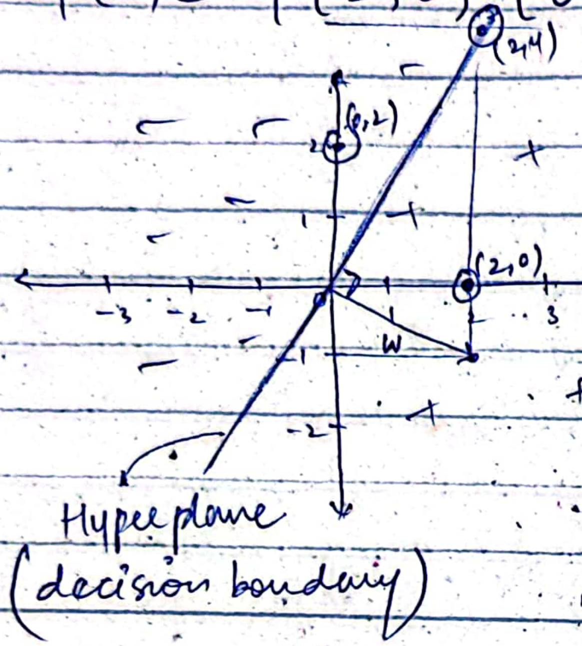
$\in \mathbb{R}$ → point

$\phi(x) \in \mathbb{R}^2$ → line (that separates +ve and -ve)

$\in \mathbb{R}^d$ → plane

$w = [2, -1]$

$\phi(x) \in \{ \underset{x_1}{[2, 0]}, \underset{x_2}{[0, 2]}, \underset{x_3}{[2, 4]} \}$



Example → angle with $\vec{w}$

Acute ∠ → +ive

Obtuse ∠ → -ive

Right ⊥ → indecisive.

$w \cdot \phi(x_1) = 4 + 0 = 4 > 0 \; (+1)$

$w \cdot \phi(x_2) = 0 - 2 = -2 < 0 \; (-1)$

$w \cdot \phi(x_3) = 4 - 4 = 0 \quad ?$

<u>Loss function</u>:

Ghalat predictions btata hai:

Prediction ≠ Ground Truth.

<u>Margine</u>: Score * y ← ground truth.

$= (w \cdot \phi(x)) * y$.

if margine < 0 → Wrong Prediction.

else → Correct Prediction.

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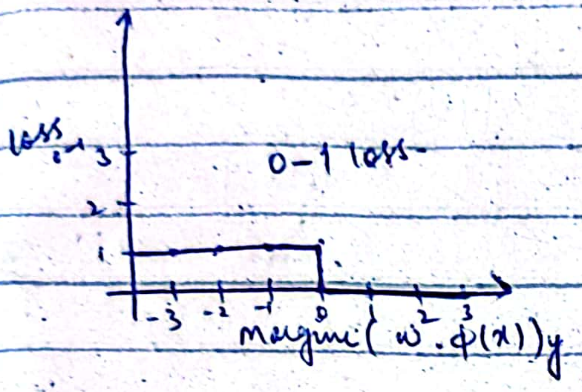
Reflex Models:-

BINARY CLASSIFICATION:

- 0-1 Loss function :-

$$\text{loss}_{0-1}(x, y, w) = 1\,[f_w(x) \neq y] \quad y' \text{ (predicted output)}, \; y \text{ — target value}$$

$$= 1\,[(w \cdot \phi(x))y \leq 0] \longrightarrow \text{loss defined using margin}$$

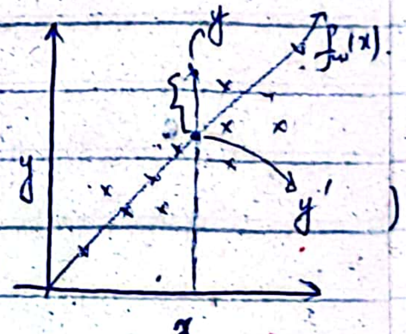


LINEAR REGRESSION:

0-1 loss can't be used

bcz $y' \neq y$ (Exactly not equal)

$y' = f_w(x) = w \cdot \phi(x) \in \mathbb{R}$

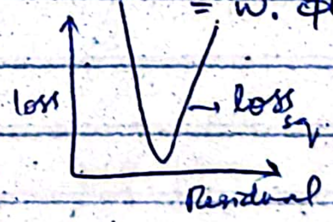


$y - y' = \text{Residual}$

$= w \cdot \phi(x)$

- Squared loss:-

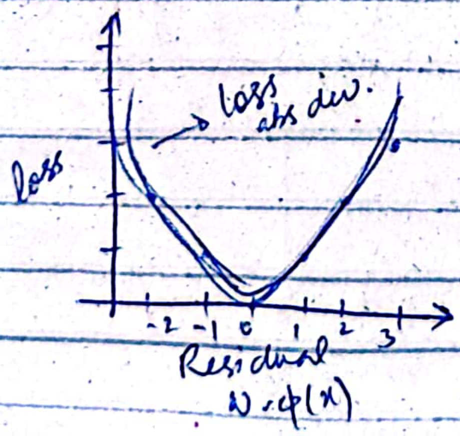
$$\text{loss}_{sq}(x, y, w) = (w \cdot \phi(x) - y)^2$$



- Absolute deviation loss function:-

$$\text{loss}_{\text{abs dev.}}(x, y, w) = |\, w \cdot \phi(x) - y \,|$$

absolute ?
Magnitude of loss can't be -ive



Goal :-

minimize loss.

Loss Minimization :-

$$\text{Train Loss}(w) = \frac{1}{|D_{train}|} \sum_{(x,y) \in D_{train}} \text{Loss}(x, y, w)$$

↑ no. of training Σx

derivative w.r.t w

$$= \frac{2}{|D_t|} \Sigma (w \cdot \phi(x) - y) \cdot \phi(x)$$

⇒ Outliers can affect in squared loss func.
not so much in Absolute Dev. loss fun.

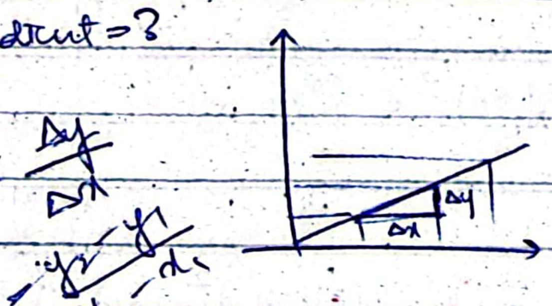
Objective

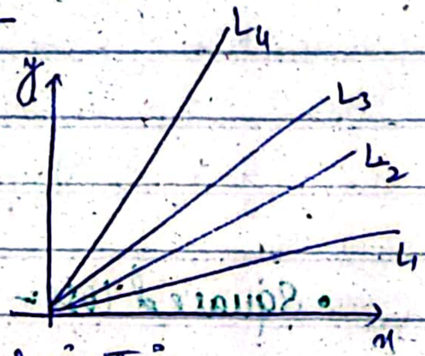
$$\min_{w \in R} \text{Trainloss}(w)$$

- **Gradient Descent Algo** :-

Gradient = ?

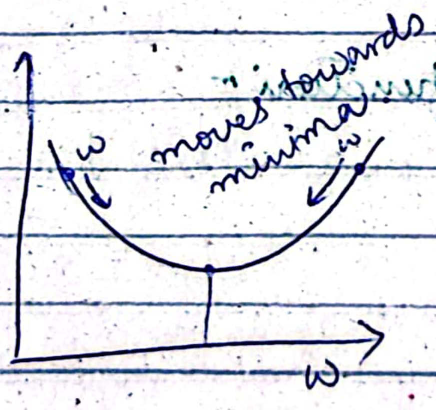
slope = $\frac{\Delta y}{\Delta x}$

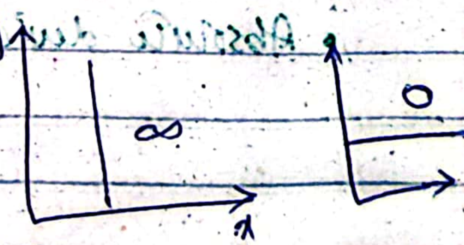




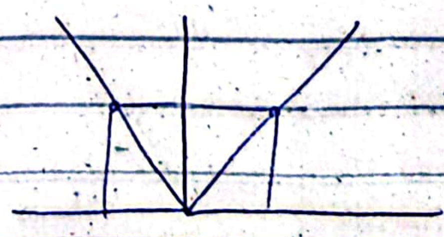
derivative

$L_4 > L_3 > L_2 > L_1$





$\nabla_w \text{trainloss}(w)$



dev. same
sign. opp.

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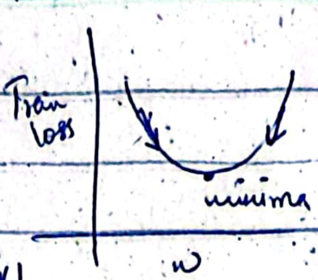
Gradient Descent Algo.



$w = [0, 0, \dots]$

for $t = 1, \dots, T$

$w = w - \eta \nabla_w \text{Trainloss}(w)$

derivative $\rightarrow 2(w \cdot \phi(x) - y)\phi(x)$

Hyper parameter :-

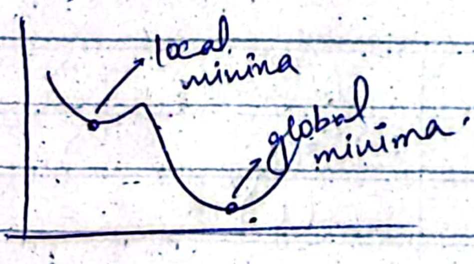
- Do not depend on algo.
- fixed experimentally.

→ T ⇒ no. of iterations

→ η ⇒ Eta → Step size → $\frac{1}{\sqrt{\text{no. of updates so far}}}$

can be fixed to 0.1 or 0.01 too.

- Step size bara ho tou, you keep oscillating around the minima.
- Step size chota ho tou it will take too long to reach minima. and you might stick to local minima, skipping the global.



→ Puray data set ka Train loss nikal k, phir ek baar w (weight) update hota hai.

→ very slow.

→ also called "Batch Gradient Descent"

So,

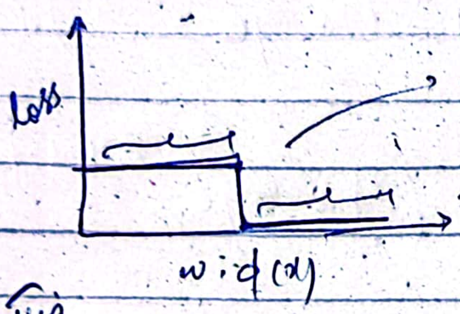
Stochastic Gradient Descent :-

- update w after every example.

e.g. Agar data set mein million examples hain, tou w million times update hoga in one iteration.

⇒ Agar Pura Data Set hai, tou Batch G.D lag sakta hai.
⇒ Agar Data Set hai hi nai, Dynamic values le rahay hain tou Stochastic G.D hi lagay ga.

0-1 loss :- (Classification Prblm)



gradient zero hai (straight line)
w wont update
Algo. wont progress.

for this → **Hinge Loss :-**

$$\text{loss}_{\text{hinge}}(x, y, w) = \max\{\overbrace{1-(w\cdot\phi(x))y}^{\text{margin}}, 0\}$$

in doono me se max.

$$\text{loss}(x,y,w) = \begin{cases} 0 \; ; & w\cdot\phi(x)y \geq 1 \\ 1-(w\cdot\phi(x))y \; ; & \text{other} \end{cases}$$

derivative :-

$$\nabla \text{loss} = \begin{cases} 0 \\ 0 - \underbrace{\phi(x)y}_{-\phi(x)\cdot y} + \underbrace{\frac{dy}{dw}}_{\text{zero}} \end{cases}$$

Logistic Loss :-

$$\text{loss}_{\text{logistic}}(x, y, w) = \log\left(1 + e^{-(w\cdot\phi(x))y}\right)$$

↳ never touches zero on graph.
↳ keeps updating values.

derivative

$$\nabla_w \text{loss}_{\text{logistic}} = 1 - \left(1 + \log e^{-(w\cdot\phi(x))y}\right) \cdot \big($$

$$\left(1 + \log e^{-(w\cdot\phi(x))y}\right)$$

Summary :-

	Classification	Regression
Prediction f_w	sign(score)	score
Prediction ?	Margin	Residual
Loss funct.	0-1 loss loss hinge logistic	Squared Abs. dev. loss.
Algo	SGD	SGD

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Hypothesis Class:

Set of predictors with fixed $\phi(x)$ and varying w.

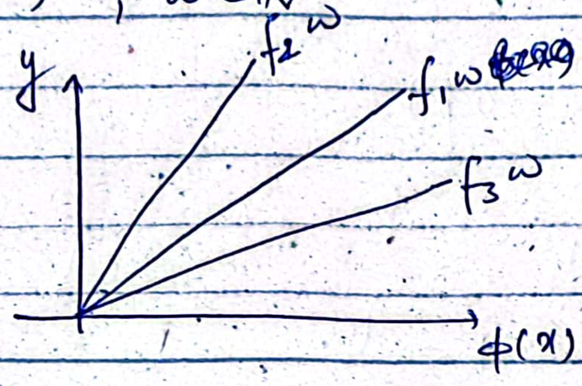
$$F = \{ f_w : w \in \mathbb{R}^d \}$$

$f_w = w \cdot \phi(x)$, $w \in \mathbb{R}$

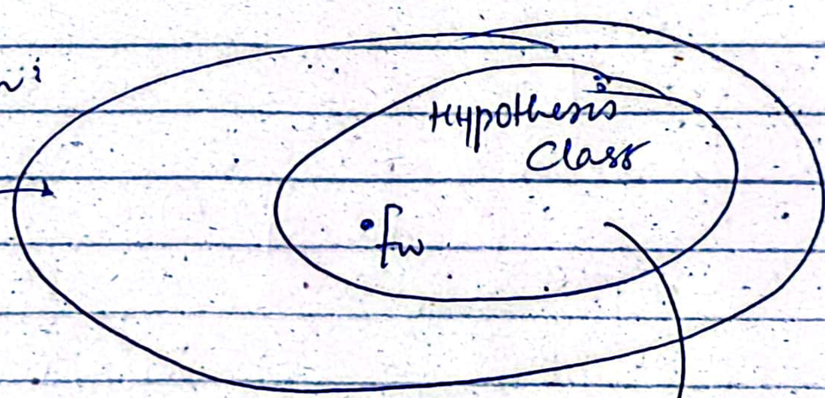
$f_1 w = \phi(x)$

$f_2 w = 2\phi(x)$

$f_3 w = 0.5\phi(x)$



Set of features given:



(Good feature selection will lead to good Hyp. Class.)

Feature Extraction!!!

Learning → using data.

↳ gives parameters (w).

of from within the set.

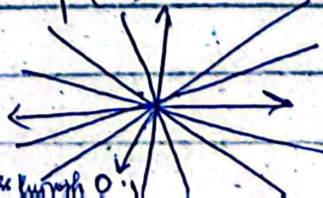
→ data should be diverse for good learning and finding best predictor.

Example:

Reg: $x \in \mathbb{R}$, $y \in \mathbb{R}$. Let $\phi(x) = x$.

$$F_1 = \{ w_1 x + w_2 x^2 : w_1 \in \mathbb{R}, w_2 = 0 \}$$

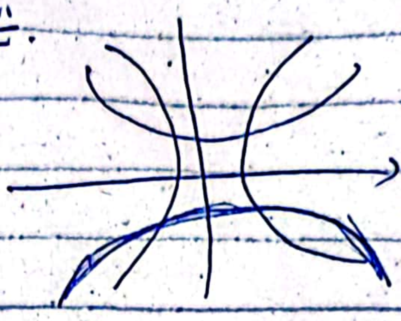
(Straight line)



$F_c = \{ w_1 x + w_2 x^2 + c \}$

$F_2 = \{ w_1 x + w_2 x^2 \; ; \; w_1 \in \mathbb{R}, w_2 \in \mathbb{R} \}$

if $w_2 \neq 0 \rightarrow$ parabola.

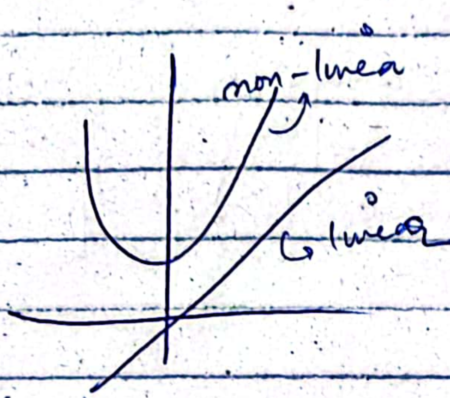


- Learning Algo. will push w_2 towards zero.!

→ linear predictors (easy)

$F_1 \subset F_2$

↓ subset.



$F_1 \subset F_c$

Score := $w \cdot \phi(x)$

→ linear in w

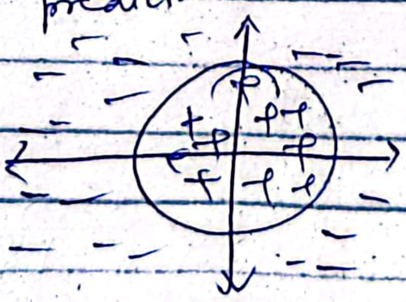
→ linear in $\phi(x)$

$f_w(x) \Rightarrow$ non-linear.

non-linear $w_2 x^2$

$x_1^2, \; x_2^2, \; b$ $\qquad x_1^2 + x_2^2 + b = 0$

↑ w_1 ↑ w_2 ↑ biased predictor ↓ circle



NON- LINEAR MODELS:-

(1.) Neutral Networks:-

↳ helps in finding good features from raw data.
   ↳ then makes prediction.

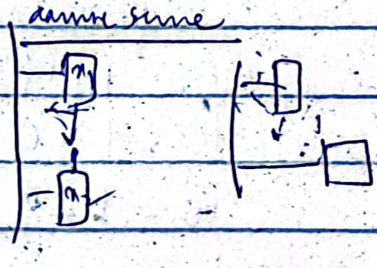
Adv :- • Burden of FE shift to NN.
   • Focus on Learning.

Motivational E.g.

→ Predicting car collisions

Input : Position of incomp. Cars $x = [x_1 \quad x_2]$

Output : Safe $(y = +1)$, Collide $(y = -1)$,



$x_1 - x_2 \geq 1$ $\quad x_1 - x_2$

$x_2 - x_1 \geq 1$ $\quad \boxed{x_2 - 3 - 1.2}$

$$y = \text{Sign}(|x_1 - x_2| - 1)$$

If car is

$$h_1 = 1[x_1 - x_2 \geq 1]$$

$$y = \text{sign}(h_1 + h_2)$$

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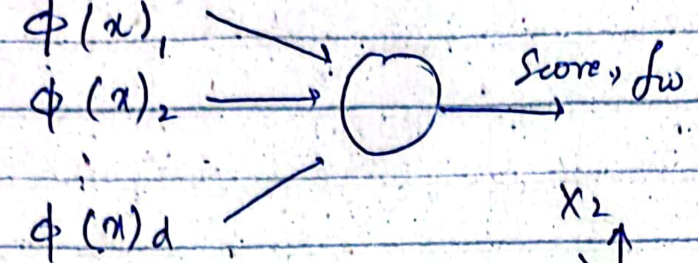
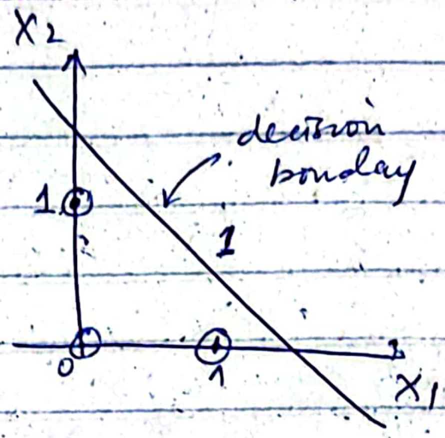
Neural Networks

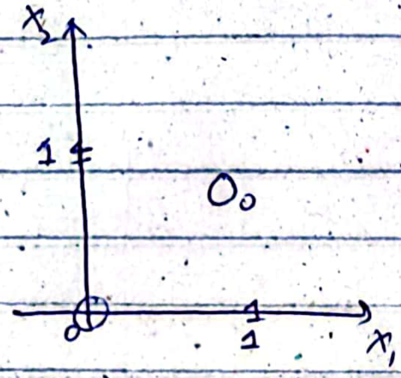
- single layer NN.



X_1 AND X_2		Y
0	0	0
0	1	0
1	0	0
1	1	1



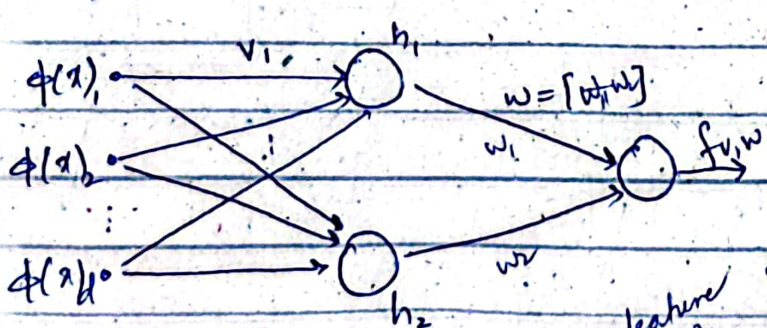
X_1 XOR X_2		Y
0	0	0
0	1	1
1	0	1
1	1	0



"can't make decision boundary." (linear)

→ Multi-layer NN:-

• 2 layer NN :-



h_1, h_2 → hidden units.

$h \in \mathbb{R}^k$

$w \in \mathbb{R}^k$

$\phi(x) \in \mathbb{R}^d$

$v_1 \in \mathbb{R}^d$

$v_2 \in \mathbb{R}^d$

hidden unit, feature no.

$v_1 = [v_{11}, v_{12}, \ldots, v_{1d}]$

no. of hidden units.

$$f_{V,w} = \sum_{j=1}^{k} w_j \underbrace{\sigma(V_j \phi(x))}_{h_j}$$

$$= w_1 \sigma(V_1 \phi(x)) + w_2 \sigma(V_2 \phi(x))$$

$$[\,—\,]_d \quad \begin{bmatrix} | \end{bmatrix}_d$$

yahan se

$$Loss(x, y, V, w) = (f_{V,w}(x) - y)^2$$

↳ for each input

yahan se

<u>Objective</u> :- Loss Minimization of Train Loss.
depends on V, w.

$$\text{TrainLoss}(V, w) = \frac{1}{|D_{train}|} \sum_{(x,y) \in D_{train}} Loss(x, y, V, w)$$

Parameters Updation :-

at the end after all summed up

$$w = w - \eta \nabla_w \text{TrainLoss}(V, w)$$
$$V = V - \eta \nabla_V \text{TrainLoss}(V, w)$$

Batch Gradient Descent

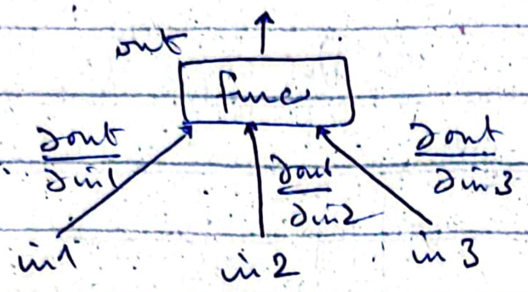
w: updated with each example

Stochastic GD

$$\nabla Loss(x, y, V, w).$$

Gradient Computation:

↳ using Computation Graphs.



Ex $2 * in1 + in2 * in3 = out$

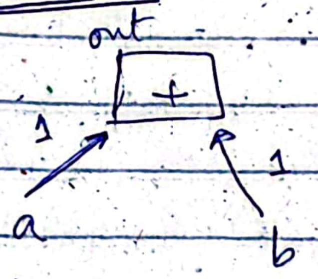
$2(in1+\epsilon) + in2 * in3 = out + \underbrace{2\epsilon}$

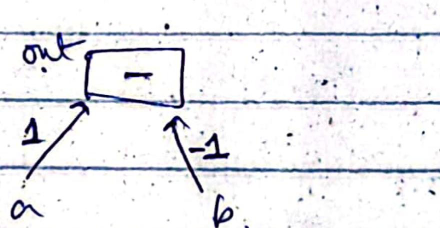
$$\frac{\partial out}{\partial in1} = \frac{2\cancel{\epsilon}}{\cancel{\epsilon}} = 2$$

$2 * in1 + (in2+\epsilon) in3 = out + \underbrace{in3 * \epsilon}$

$$\frac{\partial out}{\partial in2} = \frac{in3 * \cancel{\epsilon}}{\cancel{\epsilon}} = in3$$

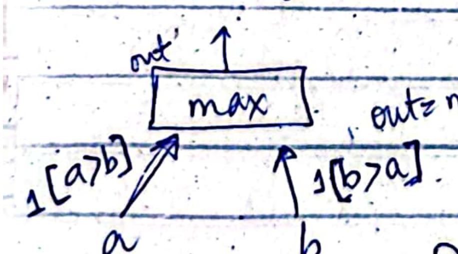
Basics of derivative:



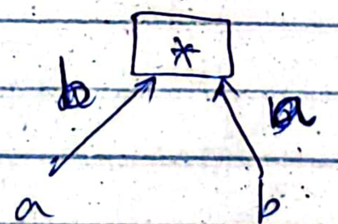


$\partial_a(a+b) = 1$

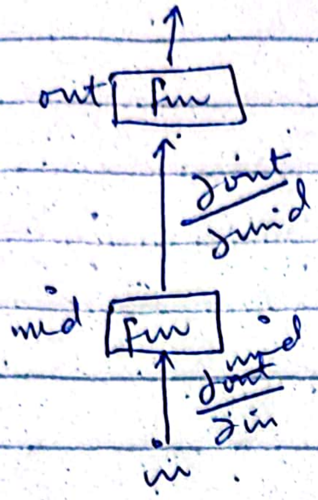
$\partial_a(a-b) = -1$



out = max(a, b)



$$\frac{\partial ab}{\partial a} \qquad \frac{\partial a\cancel{b}}{\cancel{b}}$$

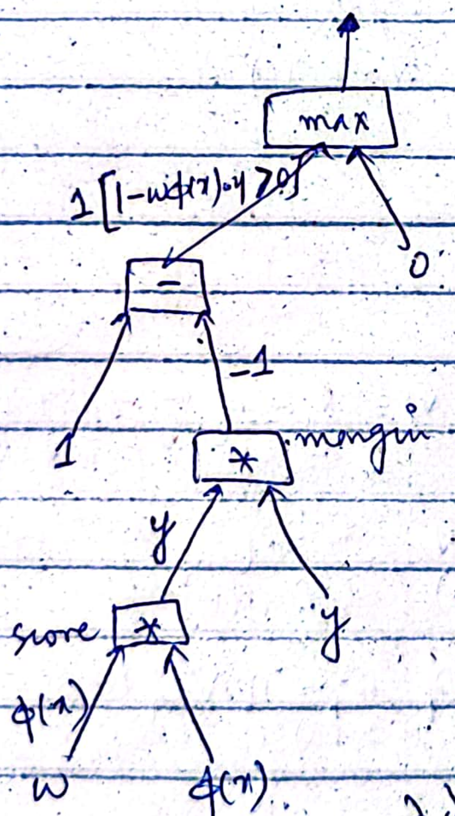


Chain Rule

$$\frac{\partial out}{\partial in} = \frac{\partial mid}{\partial in} \cdot \frac{\partial out}{\partial mid}$$

Hinge loss :-

$$loss(x, y, w) = \max\{1 - w\phi(x)\cdot y\ ,\ 0\}$$



$$\frac{\partial loss(x, y, w)}{\partial w} = ?$$

$$1[1 - score\cdot y > 0]\,(-1)(y)(\phi(x))$$

$$= [1 - margin > 0]\cdot(-1)\cdot(y)\cdot(\phi(x))$$

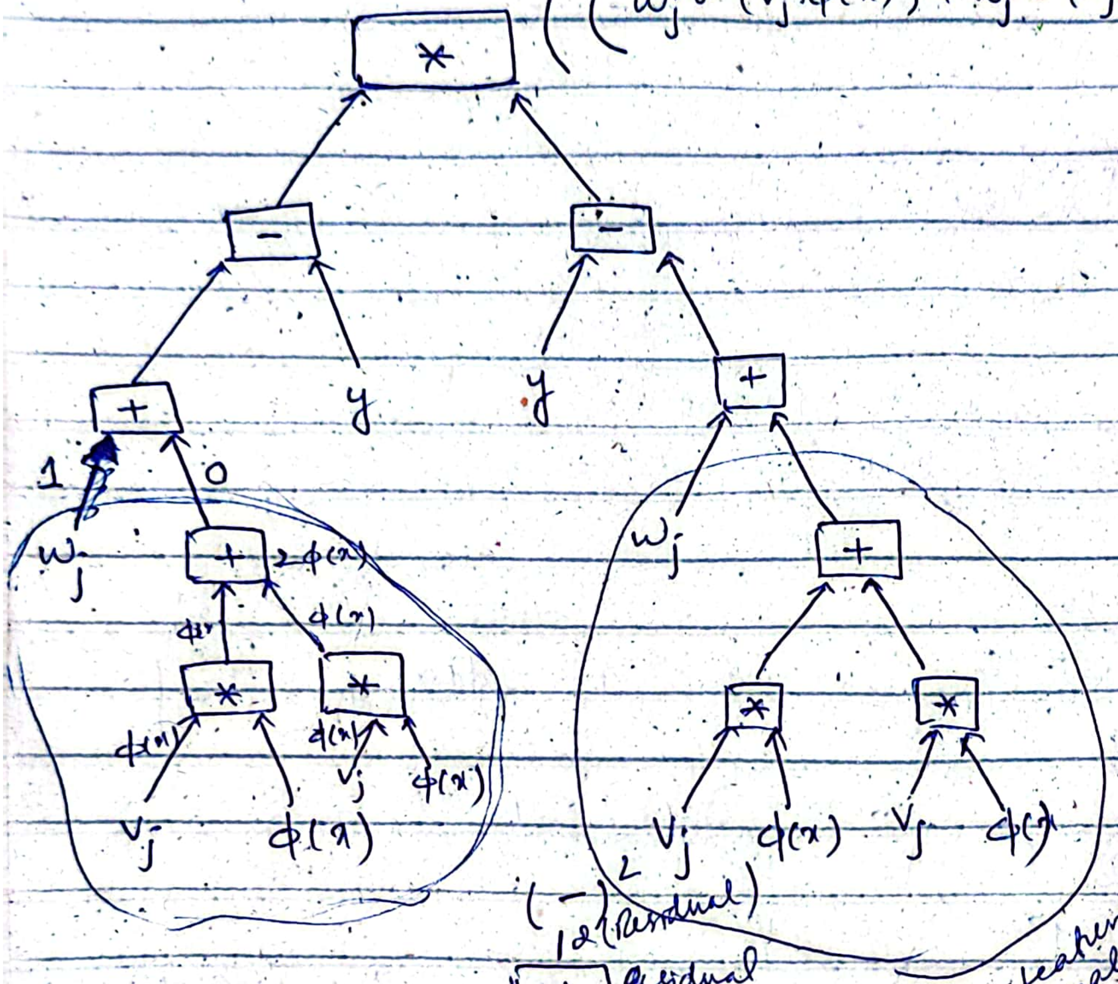
$$\frac{\partial \cancel{w}\phi(x)}{\partial \cancel{w}}$$

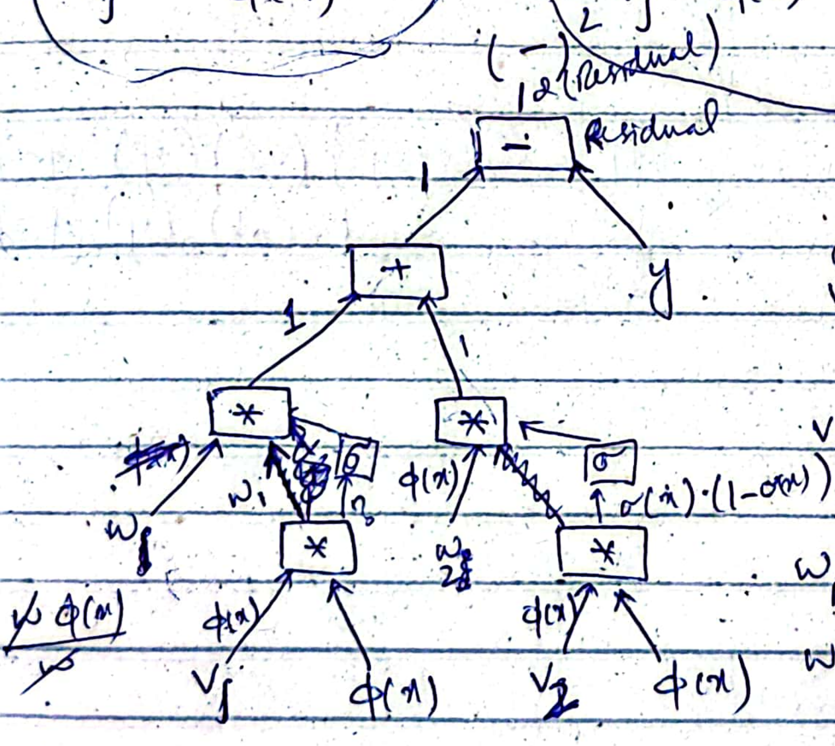
hidden layer

(assume $k=2$)

$$\text{Loss}(x, y, V, w) = \left(\sum_{j=1}^{k} w_j \, \sigma(v_j \cdot \phi(x)) - y \right)^2$$

$$\left(\left(w_j \, \sigma(v_j \cdot \phi(x)) + w_j \, \sigma(v_j \, \phi(x)) \right) - y \right)^2$$





size of feature vector each value will be updated

learning rate

$$v_1 = v_1 - \eta \nabla_{v_1} \text{Loss}(x, y, v, w)$$
$$= 2(\text{Res.})(1)(1)(\sigma_x(1-\sigma_x))(v_1)$$

$$v_2 = v_2 - \eta \nabla_{v_2} \text{Loss}(x, y, v, w)$$
$$= (2)(\text{Res.})(1)(1)(\sigma_x(1-\sigma_x))(v_2)$$

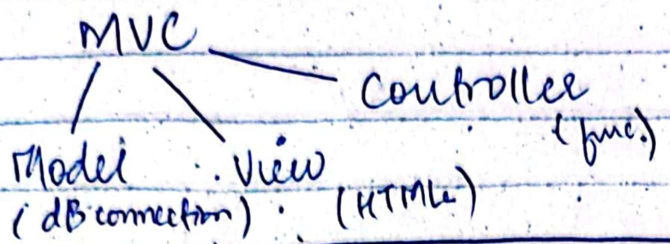
$$w_1 = w_1 - \eta \nabla_{w_1} \text{Loss}(\;)$$
$$= 2(\text{Res})(1)(1)(\sigma_x)(1-\sigma_x)$$

$$w_2 = w_2 - \eta \nabla_{w_2} \text{Loss}(\;)$$
$$= 2(\text{Res})(1)(1)(\sigma_x)(1-\sigma_x)$$

(derivative of σ) $\sigma'_x = \sigma x(1 - \sigma x)$

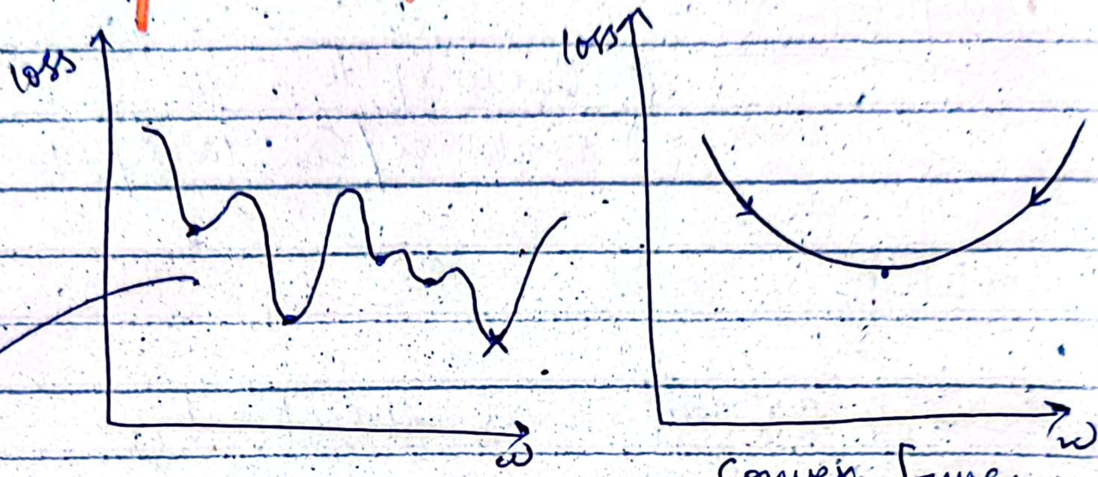
13Nov2023 CS-457 WAF

MVC Design Pattern



14Nov2023 **Optimization :-**

AI



→ Linear predictor will get stuck at a local minima in such scenario.

→ Gradient Descent might act the same way too.

↳ Deep Neural Networks escape local and eventually find the global minima.

K-NN :- (K-Nearest Neighbour)

used for
- ↳ classification
- ↳ Regression

Algo :-
1. Training : D_{train}

$k=1$
2. Input x'.
3. Predictor $f(x')$:
   - find $(x,y) \in D_{train}$

Minimum Distance

where $\|\phi(x) - \phi(x')\|$ is smallest if k = 3 → find 3 such examples that are nearest.

Classification → Then decide on voting.

Regression → Take their mean.

Imp. Things :-

① Value of k.

② Distance/Similarity Measure

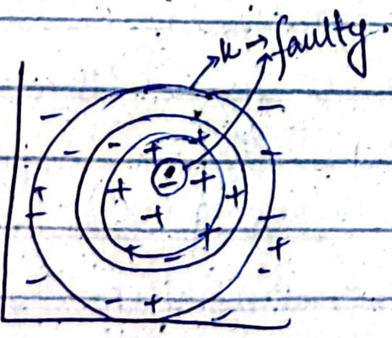
Value of k :-

→ if k is small i.e. k = 1.

Cons :- • Data may be noisy (rare)
• Misprediction chances high.
• Decision based on one e.g. ↓ that might be faulty.

→ if k is large i.e.

Cons :- • Might include other class region.

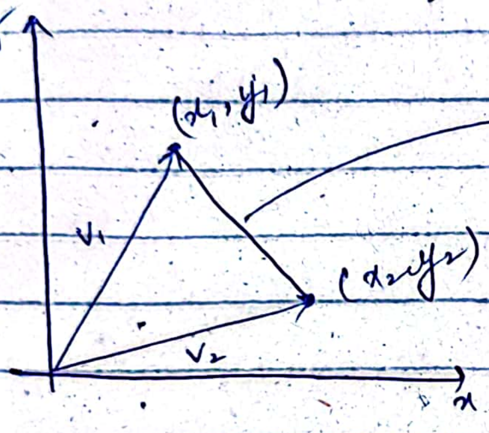


2 examples → distance less / distance more
↓ ↓
more similar less similar.

Distance/Similarity Measures :-

Vector Norms :-

$V_1(x_1, y_1)$, $V_2(x_2, y_2)$



→ L_2 Norm / Euclidean Distance

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

L_1 Norm / Manhatten Distance

$$D = |x_2 - x_1| + |y_2 - y_1|$$

Infinity Norm :-

$$D = \max(|x_2 - x_1|, |y_2 - y_1|)$$

Example:

$V_1(2, 7)$
$V_2(12, 1)$

L2-Norm :- $\sqrt{(12-2)^2 + (1-7)^2} = 11.66$ (max. distance gets more weightage)

L1-Norm :- $|12-2| + |1-7| = 16$ (both get equal weightage)

Infinity :- $(|12-2|, |1-7|) = 10$ (max. distance gets all the weightage) features

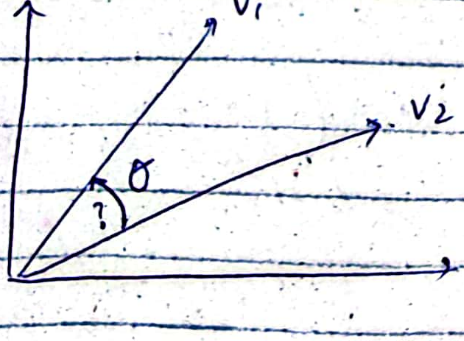
$A = \langle 5, 5, 5 \rangle$
$B = \langle 2, 2, 2 \rangle$
$C = \langle 3, 2, 5 \rangle$

1-Norm: $(A, B) = |2-5| + |2-5| + |2-5|$
$= 3 + 3 + 3 = 9$
$(B, C) = |3-5| + |2-5| + |5-5|$
$= 2 + 3 + 0 = 5$

$\vec{B}$ and $\vec{C}$ seem more similar based on Norm (Distance).

$\vec{A}$ and $\vec{B}$ seem more similar based on features' composition. (same feature).

To find this, we have a measure
(<u>Cosine Similarity</u>)



<u>Feature Normalization</u>:
(0-1)

$$X_n = \frac{X - min}{max - min}$$

13/11/2023

Generalization :-

→ Obj. func.

$$\text{Train loss}(w) = \frac{1}{|D_{train}|} \sum_{(x,y) \in D_{train}} \text{loss}(x, y, w)$$

→ is it a True Obj. ?
Unseen future examples pe predictor sahi kaam karay.

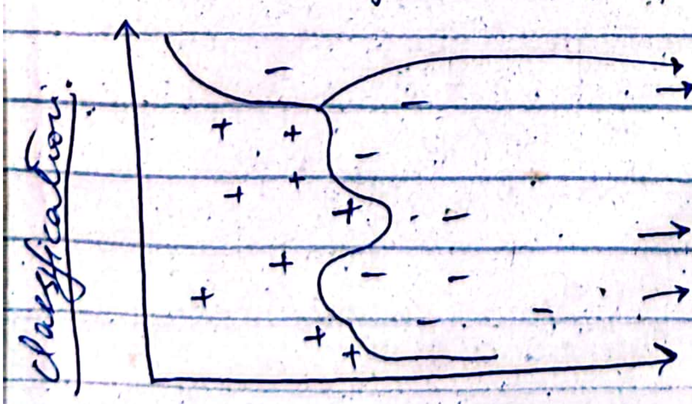
Algos. (Strawman Approach)
Training Data : Store Dtrain
Predictor f(x) :
if (x,y) ∈ Dtrain return y..
else error.

Not our Objective.

→ is case me error/loss "zero" hoga.

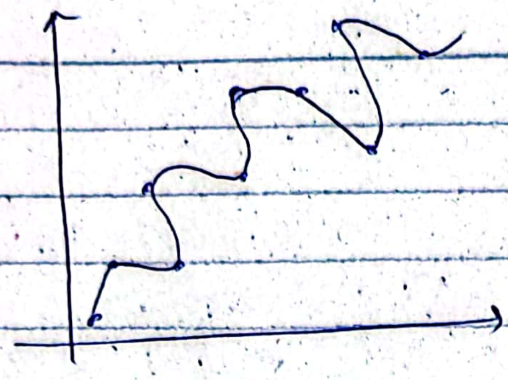
→ Agar Train.loss tou minimum hai, par future examples me accuracy achi nai hai, tou hum kehte hain k algos. Generalized nai hai.

So, Overfitting and Underfitting :-



→ 100% accurate decision boundary. (error ≃ 0)
→ Not generalized.
→ Overfitting / Overtraining
(Memorises Training Data)

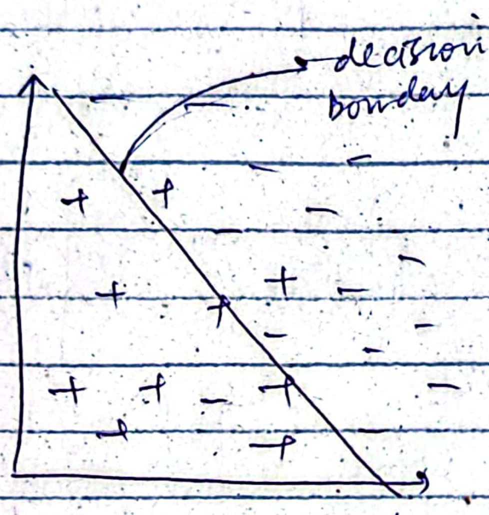
Regression



→ Overfitted / Overtraining
→ 100% accuracy
→ Training Error ≃ 0.

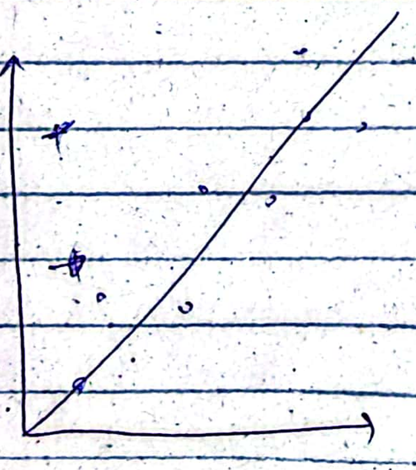
On unseen data
error >> 0.

Classification



→ Underfitting.

→ Simple Algo.
on complex prblm.



→ Underfitting.

→ Train Err >> 0
future error >> 0.

⇛ How to Solve This Problem!?

$D_{train} \longrightarrow \boxed{\text{Learner}} \longrightarrow f$

How good is the predictor f ???

① Split the Data Set. (Train and Test)

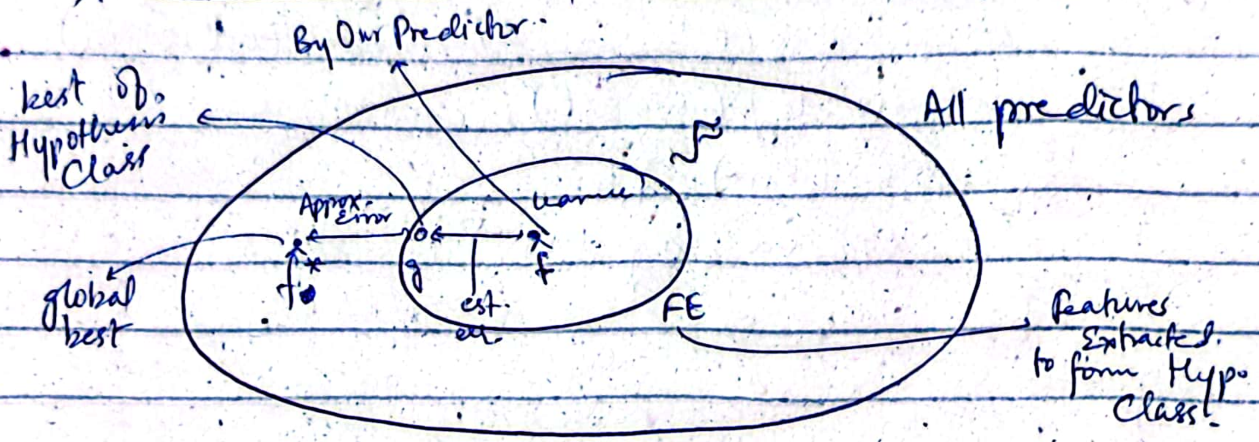
Train: seen by model
Test: unseen. like future data.

Note :-

- The more the Dtrain , the more diversity will be captured.
- Incase of very large Data Large Data Sets, no need to put even 10%. data for test.

Errors :- Approx. Error + Est. Error = Error (Overall)

$\Rightarrow$ Approx. Error & Estimation Error



Est. Error :- How good is the learnt predictor relative to the potential of Hypothesis Class.

Approx. Error :- How good is the Hypothesis Class? relative to the

→ Increase Data. features have the potential.

NOTE :-

Optimum Sol. is not known in ML.
So, we Tryna go towards Near best Sol.

21/11/2023

(CS-410)
AI

Size of Hypothesis Class :-

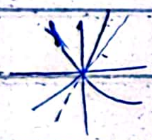
Increase the size of Hyp. Class
↳ • Approx. Error decreases.
↳ more capability of reaching close to the global minima. (f^*)

• Est. Error increases.
↳ search space berhay gi.
↳ near best away ki chances hard.

⇒ How to control the size of Hyp. Class?:

① ↳ "dimensionality" ko reduce keena.

$F_1 = \{ f_w = w_1 x + w_2 x^2 \;;\; w_1 \in \mathbb{R},\ w_2 = 0 \}$ ← linear predic



$F_2 = \{ \_\_\_\_ \;;\; w_1 \in \mathbb{R},\ w_2 \in \mathbb{R} \}$ ← linear + quadratic.

$F_1 \subseteq F_2$

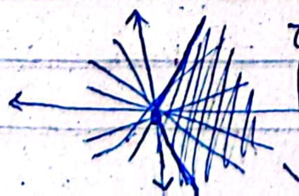
↳ features woh nikalne hain jo ya accuracy ko kam kar rahay hain ya help nai kar rahay in good prediction.
⇒ Exclude irrelavent ones only.

② Controlling the "Norm" :-

$w \in \mathbb{R}^d$

$\|w\| = \sqrt{(w_1)^2 + (w_2)^2 + \dots + (w_d)^2}$

↳ Reduce the length (norm) $\|w\|$.



} reduce region

⇒ Regulization

penalty on Error

$\boxed{\min \text{Train loss}(w) + \frac{\lambda}{2} \|w\|^2}$

$= \frac{\lambda}{2}||w||^2$

$= 2w \cdot \frac{\lambda}{2}$

$= \lambda w$

Don't increase the weights. $\rightarrow$ $w = w - \eta \nabla_w \{trainloss + \frac{\lambda}{2}||w||^2\}$

Penalty zyada hogi verna.

↳ Early Stopping :-

bari iteration u kuje chalaya

ton Over-training hogi. Not good.

– So stop early.

↳ make T smaller

$T \downarrow$ no. of iterations.

Hyper-Parameters :-

$T \rightarrow$ no. of Iterations

$\lambda \rightarrow$ Regularization Parameter

↳ (0–1) Range

$nf \rightarrow$ no. of features.

⇒ How to choose these?

Choose to minimize D_{train} error ? NO!

↳ to do this

$\lambda = 0$, $T = max(\infty)$, $nf = all$.

↳ Results in Overfitting

choose Go to minimize Dtest Error? No!

↳ Tuned parameters by all data.

So it is no more a Test Data.

Solution:

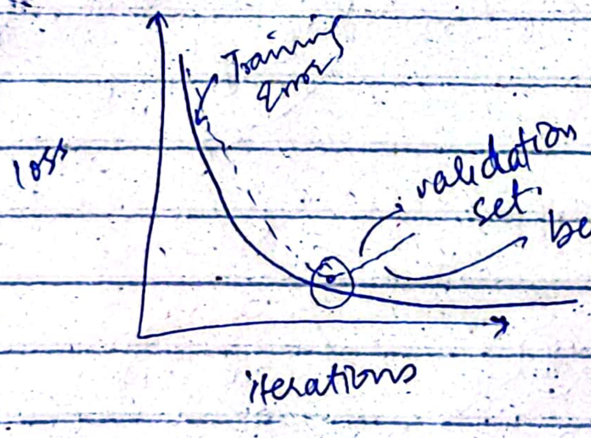
Divide DataSet in 3 parts.

Training — what should be Optimal size of Train set??

~~Training~~ ~~Validation~~

Validation ↑ for HyperParameter tuning.

Test (not to be shown at any cost)



bachney lag jayega! at a point

↓

Time to Stop your T.

24/11/2023

CS-414
AI

Control The Norm :-

① **Regularization :-**

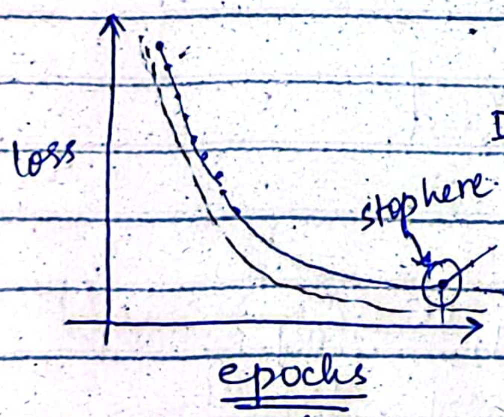
$$\min_{w} \text{ Trainloss}(w) + \boxed{\frac{\lambda}{2} \|w\|^2}$$ penalty

Value of λ :-

too big → not focused on Loss.
too small → not focus on norm.

② **Early Stopping :-**

Don't overfit.



check error on this set

D_{train} , $\boxed{D_{val}}$, D_{test}

early stop here:
issue baad agay gaye tou error increase hoga.

epochs → sara data de diya, loss compute kia } first epoch
give all the data again and compute loss } for next epoch.

NOTE :-

→ Validation Set k case me error Training Set ki nisbat zyada hi hoga.

↳ cz D_{train} pe hum w (weights) ko update kerte hain, D_{val} me nai kerte.

- Imbalanced data causes biased results.

Supervised learning → $D_{train}(x, y)$
Input Output
(Labelled Data)

Unsupervised Learning:-

We don't know the Target.

→ D_{train} contains only Imp. features.
→ Unlabelled data.
→ Labelled data is expensive in Real World. and you can't rely on one resource.

1- Clustering.
2- Dimentionality Reduction (PCA).

① Clustering:-

1- Word Clustering.

Input : Raw words/texts
Output : Clusters.

Cluster 1 : Monday, Friday, Sunday
Cluster 2 : January, February, December.
Cluster 3 : Apple, Mango, Oranges.
Cluster 4 : Mother, Father, Brother.

Word Vectors:- (i.e we can't manipulate txt).

Vector Space Representation

1) One hot Representation.

↳ Vector size will be total words count in the lib/dictionary.

Only that specific word will be 1, rest all 0.

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

28/11/2023 CS-474 / AI

Vector Space Representation!

→ One Hot Rep.
→ Word2vec:

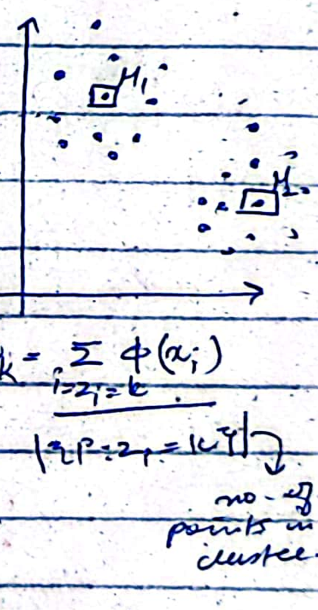
$$\begin{bmatrix} \vdots \\ \text{Monday} \\ \vdots \\ \text{weekday} \\ \vdots \end{bmatrix} \quad \begin{bmatrix} \vdots \\ 1 \\ \vdots \\ 0.9 \\ \vdots \end{bmatrix}$$

K-Means Algo.:- (~~K-Mea~~ Clustering)

→ Objective: Similar points in one cluster.
→ Input: D_{train}.
→ Output: Assign each point to a cluster.

Set up:-

1) Each cluster $1, 2, \ldots, K$ is represented by a Centroid $\mu_K \in \mathbb{R}^d$ → dimensions / features
2) Each point $\phi(x_i)$ close to its assigned Centroid (μ_z).



Obj. funct.

$$Loss(z, \mu) = \sum_{i=1}^{n} \| \phi(x_i) - \mu_{z_i} \|^2$$

$$z_i = \arg\min_{k=1,2,3..} \| \phi(x_i) - \mu_k \|^2$$

$$\mu_k = \frac{\sum_{i: z_i = k} \phi(x_i)}{|\{i : z_i = k\}|}$$ → no. of points in cluster.

Eg:-

$D_{train} = \{0, 2, 10, 12\}$

Assume $\mu_1 = 1, \ \mu_2 = 11$

assignment:

$z_1 = \arg\min \{ (0-1)^2, (0-11)^2 \} = 1$

$z_2 = \arg\min \{ (2-1)^2, (2-11)^2 \} = 1$

$z_3 = \ " \ \ " \ \{ (10-1)^2, (10-11)^2 \} = 2$

$z_4 = \ " \ \ " \ \{ (12-1)^2, (12-11)^2 \} = 2$

Given zs can compute Centroid,

$\mu_1 = ?, \ \mu_2 = ?$

$\mu_1 = \frac{0+2}{2} = 1 \qquad \mu_2 = \frac{10+12}{2} = 11.$

Stop iteration when they don't change

Input:- $k = 2$.

$D_{train} = \{ (10,0), (9,6), (8,2), (10,7), (9,0) \}$

$\mu_1 = (10,0)$ $\quad\quad$ $\mu_2 = (8,2)$.

$z_1 = \sqrt{(10-10)^2 + (0-0)^2} = \cancel{8}\ 0$ ✓ 1

$\sqrt{(10-8)^2 + (0-2)^2} = \cancel{\sqrt{8}}\ \sqrt{8}$

$z_2 = \sqrt{(9-10)^2 + (6-0)^2} = \sqrt{37}$ $\quad$ 2

$\sqrt{(9-8)^2 + (6-2)^2} = \sqrt{17}$ ✓

$z_3 = \sqrt{(8-10)^2 + \cancel{8}(2-0)^2} = \sqrt{8}$

$= \sqrt{(8-8)^2 + (2-2)^2} = 0$ ✓ $\quad$ 2

$z_4 = \sqrt{(10-10)^2 + (7-0)^2} = \cancel{9}\ 7$ ✓ $\quad$ 1

$\sqrt{(10-8)^2 + (7-2)^2} = \cancel{50}\ \sqrt{29}$

$z_5 = \sqrt{(9-10)^2 + (0-0)^2} = \sqrt{1}$ ✓ $\quad$ 1

$\sqrt{(9-8)^2 + (0-2)^2} = \sqrt{5}$

$\mu_1 = (10,0) + (10,7) + (9,0)$

$= \left(\frac{29}{3}, \frac{7}{3}\right) = (9.6, 2.3)$

$\mu_2 = (9,6) + (8,2)$

$= \left(\frac{17}{2}, \frac{8}{2}\right) = (8.5, 4)$